



NETWORK DESIGN TEAM

(325)

FINAL

STATE 2022

Description

Analyze existing and planned business environments and develop a strategy for the implementation of a network infrastructure that addresses the business needs of the scenario provided.

Topic

The world of esports is a massively connected multi-billion-dollar industry. DF Solutions, a technology support contractor for small, medium, and large-scale conference/special events, has recently decided to enter the world of esports event planning. Your team's task is to create a network infrastructure plan (replicable models) to accommodate Regional, State, and National esports conferences based on the information below.

- Local or Regional esports Conference:
 - This model should define the requirements for creating a small-scale network to accommodate up to 200 players.
- State esports Conference:
 - This model should define the requirements for creating a mid-scale network to accommodate up to 1000 players.
- National esports Conference:
 - This model should define the requirements for creating a large-scale network to accommodate up to 5000 players.

Requirements to consider for all levels:

- Bandwidth requirements to accommodate low-latency gameplay
- Secure storage (network shared) for all games
- Anti-cheat systems
- Closed vs open network design
- Equipment requirements (computers, routers, switches, etc.)

Teams who do *not* submit an entry following this topic will be *disqualified*.

Modification for State Final:

The National esports Conference needs to be modified to accommodate virtual competitors. What changes or suggestions would you make to your network design to accommodate competitors who are not able to physically attend the esports Conference.